

Winny Dariska Domkem Kameni

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Languages: English, French, German (Fluent)

PROFESSIONAL SUMMARY

Optical Test Engineer with hands-on experience in optical device characterization, automated test development, and validation within research and semiconductor R&D environments. Experienced in testing high-speed optical communication systems using Optical Spectrum Analyzers (OSA), Optical Power Meters, Source Measurement Units (SMUs), oscilloscopes, Variable Optical Attenuators (VOAs), and fiber-optic instrumentation. Proficient in developing Python-based automated test solutions using instrument APIs, with additional experience in MATLAB, LabVIEW, C, and C++. Strong background in photonics, optical communications, free-space optics, semiconductor testing, and experimental data analysis. Recognized for academic excellence with the Carleton University Senate Medal and committed to developing reliable optical test solutions from research through product validation.

TECHNICAL SKILLS

- **Optical Test & Measurement:** Optical Spectrum Analyzer (OSA), Optical Power Meter, Source Measurement Unit (SMU), Oscilloscope, Variable Optical Attenuator (VOA), Optical Fibers, Fiber Splicer, Photodiodes, Lasers, Free-Space Optical Systems
 - **Programming & Automation:** Python (Test Automation & Instrument APIs), MATLAB, LabVIEW, C, C++, Git, Data Analysis
 - **Optical & Semiconductor Engineering:** Optical Characterization, Optical Communications, Semiconductor Testing, Optical Device Validation, Experimental Design, Signal Processing, Failure Investigation, Root Cause Analysis, Test Automation, Optical Alignment, Fiber Optics
 - **Laboratory & Engineering:** R&D Testing, Hardware Debugging, Technical Documentation, Statistical Data Analysis, Experimental Validation, Cross-functional Collaboration
 - **Design Tools:** Zemax OpticStudio, SolidWorks, AutoCAD, MATLAB, Python (Automated Data Analysis).
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TECHNICAL INDUSTRY EXPERIENCE

Optical Test Technician

Credo Semiconductor — Ottawa, ON

April 2026 – Present

- Perform characterization, validation, and performance testing of optical devices supporting next-generation high-speed optical communication technologies.
- Develop and maintain Python-based automated test scripts using instrument APIs to improve test efficiency, repeatability, and data collection.
- Execute both automated and manual laboratory measurements using Optical Spectrum Analyzers (OSA), Optical Power Meters, Source Measurement Units (SMUs), and oscilloscopes.
- Validate manufacturer test data against internal engineering specifications to verify product performance and measurement accuracy.

- Troubleshoot automated test procedures, laboratory instrumentation, and measurement setups to ensure reliable and repeatable test results.
- Analyze optical measurement data with multidisciplinary engineering teams to support product development and performance evaluation.
- Document test procedures, experimental results, and engineering observations within an R&D environment.

Student Researcher

National Research Council Canada — Ottawa, ON

September 2025 – June 2026

- Develop and evaluate a MEMS-based laser beam stabilization system for free-space optical communication.
- Design and execute experiments involving fiber optics, lasers, optical alignment, photodiodes, and optical communication systems.
- Develop analysis tools using MATLAB and Python to process experimental data and evaluate system performance.
- Support optical system integration, laboratory testing, and experimental validation.
- Prepare technical documentation and present research findings to project collaborators.

Research Assistant

Carleton Optical Innovation Laboratory — Ottawa, ON

October 2024 – June 2025

- Performed optical characterization experiments using Optical Spectrum Analyzers, Optical Power Meters, Variable Optical Attenuators, oscilloscopes, optical fibers, and fiber splicing equipment.
- Assisted with optical system integration, alignment, and laboratory testing.
- Conducted experimental measurements, analyzed optical performance, and documented research results.
- Collaborated with graduate researchers on photonics and optical engineering projects.

Research Assistant

Carleton University – Systems & Computer Engineering Department — Ottawa, ON

September 2024 – January 2026

- Developed signal processing algorithms for biomedical ECG denoising using MATLAB.
- Designed digital filtering techniques to improve signal quality.
- Analyzed experimental datasets and validated algorithm performance.
- Collaborated with multidisciplinary research teams.

Teaching Assistant

Carleton University — Ottawa, ON

2024 and 2025 4 semesters

- Assisted undergraduate laboratories in C, C++ programming, and Matlab in the digital signal processing course
- Supported debugging, algorithm development, and software engineering fundamentals.

Technical Support Specialist

Xplore — Ottawa, ON

July 2023 - April 2024

- Diagnosed fiber optic and satellite communication issues.
 - Resolved technical faults through systematic troubleshooting.
 - Supported customers using telecommunications and networking technologies.
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EDUCATION

Master of Engineering (M.Eng.)

Carleton University | Ottawa, Canada | **Electrical & Computer Engineering** | **Expected Dec. 2027**

- **Relevant Focus:** Optical Communications, Photonics, Semiconductor Devices, High-Speed Systems, Test Engineering

Bachelor of Information Technology (BIT)

Carleton University | Ottawa, Canada | **Optical Systems & Sensors Engineering** | **2022- 2026**

- **Academic Standing:** GPA 11.88/12 (A+ Standing); Dean's Honour List (2023, 2024, 2025), Graduated with High Distinction, Carleton University Senate Medal.
- **Awards:** IEEE Canadian Foundation Scholarship, Amplify Immersion Scholarship, Claude Bissell Scholarship, multiple Merit Awards.
- **Coursework:** Optical Communication Networks, Digital Signal Processing, Deep Learning, Circuits & Signals, Semiconductor Physics.
- **Relevant Focus:** Photonics & Optoelectronics, Optical Communications, Optical Sensors & Instrumentation, Semiconductor Test Engineering, Fiber-Optic Systems, Signal Processing.

Advanced Diploma

Algonquin College | Ottawa, Canada | **Photonics and Laser Technology** | **2022- 2026**

- **CGPA 11.98/12**
- **Relevant Focus:** Laser Systems, Photonics & Optoelectronics, Fiber-Optic Communications, Optical Instrumentation, Optical Measurement & Testing, Optical System Integration

Bachelor of Engineering

Technical University of Applied Sciences Würzburg-Schweinfurt (THWS) | schweinfurt, Germany | **Business & Engineering** | **2022**

- **Relevant Focus:** Laser Systems, Photonics & Optoelectronics, Fiber-Optic Communications, Optical Instrumentation, Optical Measurement & Testing, Optical System Integration

PROJECT HIGHLIGHTS

National Research Council Canada (NRC) | Research Student (Optical Communication) | Sept 2025 – Present

- **Led optical link reliability modeling** for Free-Space Optical (FSO) systems, simulating atmospheric turbulence effects and optimizing transceiver alignment.
- **Performed comprehensive system-level analyses**, translating complex physical data into actionable design specifications for rural connectivity hardware.
- **Collaborated with cross-functional teams** to validate optical performance, serving as the technical lead for hardware characterization and alignment protocols.

Deep Learning Research | Lead Developer – ECG-Based Cardiovascular Detection | 2025

- **"Sold" the value of AI-based systems** to academic committees, winning multiple scholarships and merit awards based on the ability to communicate high-level impact.
- **Collaborated with global researchers**, utilizing multilingual skills (German/French) to navigate diverse technical requirements.

Carleton University | Research Assistant (Biomedical Signal Processing) | Sept 2024 – Present

- Designed **AI-based cardiovascular classifiers** and integrated wearable ECG sensors with microcontrollers.
- Implemented **Wiener and wavelet filters** to improve SNR and signal classification accuracy.

Carleton Optical Innovation Lab | Research Assistant (FBG Systems) | Sept 2024 – Apr 2025

- **Optimized multi-lens assemblies** and telescopes in Zemax, managing focal constraints and minimizing aberrations for high-precision sensing.
- **Characterized 3D-printed aspheric lenses**, measuring coupling efficiency and comparing experimental results against simulation models to refine fabrication processes.

Deep Learning Research | Lead Developer – Signal Classification | 2025

- **Built predictive models** for hardware performance, demonstrating the ability to use AI to forecast mechanical degradation and failure points.
- **Optimized signal processing** for noisy environments, a critical skill for battery testing where electrical noise is prevalent.

CERTIFICATIONS & MEMBERSHIPS

Certifications:

- TCPS 2: Ethical Conduct for Research Involving Humans
- Optica Amplify Leadership Program
- Optica Amplify Immersion Program
- IEEE Canada Student Congress (2023 & 2024)

Professional Memberships:

Optica
IEEE
Golden Key International Honour Society